

From the photon mirage to the proton mass: A pedagogical approach to the trace anomaly and confinement

Laurent Schoeffel[†]

Irfu, CEA, Université Paris-Saclay, 91191 Gif-sur-Yvette, France

[†] laurent.olivier.schoeffel@cern.ch

March 20, 2026

Abstract

The origin of mass in gauge theories is a subtle interplay between kinematics, quantum anomalies, and the macroscopic structure of spacetime. While the classical energy-momentum tensor dictates a vanishing mass for scale-invariant theories, quantum fluctuations explicitly break this symmetry via the trace anomaly. In this paper, we present a rigorous, pedagogical derivation of the mass gap equation. We explicitly demonstrate why the trace anomaly evaluates to zero for free asymptotic states in Quantum Electrodynamics (QED), ensuring a strictly massless photon even in the presence of non-linear light-by-light scattering. In stark contrast, color confinement and non-linear gluon recombination (saturation) in Quantum Chromodynamics (QCD) force the anomaly to condense, generating a finite macroscopic mass scale Λ_{QCD} for hadrons. Finally, we address the apparent paradox of energy conservation in an expanding universe, demonstrating how local Poincaré symmetry protects the microscopic quantum derivation despite the cosmological breakdown of global time-translation invariance.

Contents

1	Introduction	2
2	The mass of a quantum field: a step-by-step construction	3
2.1	The intuitive definition: rest energy and inertial mass	3
2.2	Plane waves and the local-to-global relation	4
2.3	Relativistic normalization and general tensor structure	4
3	The photon case and the trace anomaly	6
3.1	Scale invariance and the classical trace	6
3.2	Dimensional regularization and the arbitrary scale μ	6

3.3	The renormalization constant Z_e	7
3.4	Independence of the Bare Coupling and the β Function	7
3.5	Renormalization and the emergence of the anomaly	8
3.6	Evaluating the anomaly: normal ordering and the field operator	8
3.7	Matrix element computation	9
3.8	Transversality and the on-shell condition	10
3.9	The mass gap equation for the photon	10
3.10	The deep structure of QED: the Ward identity	10
3.11	Non-linear QED: the Light-by-Light contribution	11
4	QCD and the origin of the proton mass	12
4.1	The QCD Lagrangian and the chiral limit	12
4.2	The failure of the plane wave approximation	13
4.3	The QED Landau pole	14
4.4	Dimensional transmutation and the 1 GeV scale	14
4.5	The three ways of a Gauge Theory	15
4.6	The rigidity of gauge symmetry and proton structure	15
4.7	Non-linear QCD: gluon recombination and saturation	15
5	Discussion: Cosmology, curved spacetime and energy conservation	16
5.1	The cosmological breakdown of global energy	16
5.2	Local Poincaré symmetry and Riemann normal coordinates	17
5.3	Separation of scales and the modified trace anomaly	18
6	Conclusion	18
	References	19

1 Introduction

Understanding the fundamental origin of the proton mass is one of the most profound lessons drawn from the Standard Model, with direct implications for experiments at the Large Hadron Collider (LHC). In classical physics, the mass of a composite object is largely the sum of its constituents. However, in Quantum Chromodynamics (QCD), the Higgs mechanism accounts for only about 1% of the proton mass [1]. The remaining 99% emerges dynamically from the strong interaction.

This paper adopts a pedagogical approach to bridge the gap between the classical definition of mass and its quantum field theory (QFT) origin. We will explore how the energy-momentum tensor connects to the trace anomaly, contrast the behavior of the photon in QED with the gluon in QCD, and resolve the apparent tension between local

quantum dynamics and the macroscopic expansion of the universe. A central theme of this exploration is the essential role of non-linear field dynamics. We will show that while extreme non-linearities in QED (such as photon-photon scattering) cannot generate mass for a single quantum state, the inherent non-linear self-interactions of gluons in QCD are the very engine of mass generation. These collective dynamics not only confine the quarks but also mathematically stabilize the proton structure.

The purpose of these notes is to provide a rigorous, step-by-step roadmap through these concepts. In Section 2, we establish the fundamental quantum-mechanical link between the macroscopic mass of a state and the local evaluation of the energy-momentum tensor. Section 3 is devoted to Quantum Electrodynamics (QED), where we explicitly evaluate the trace anomaly on free photon states and demonstrate why the photon remains strictly massless even in the non-linear macroscopic regime. In Section 4, we turn to QCD, showing how asymptotic freedom, color confinement, and non-linear gluon saturation force the trace anomaly to condense and dynamically generate the proton mass. Finally, in Section 5, we address a profound apparent paradox: how can our microscopic derivation, which relies heavily on global energy conservation, survive in the macroscopic background of an expanding cosmological universe where time-translation invariance is explicitly broken? We resolve this tension mathematically by decoupling the fundamental physical scales and invoking the Equivalence Principle via Riemann normal coordinates.

2 The mass of a quantum field: a step-by-step construction

2.1 The intuitive definition: rest energy and inertial mass

In field theory, all dynamics (energy, momentum, and stresses) are encoded in the energy-momentum tensor $T^{\mu\nu}$ [2]. Classically, the four-momentum vector P^μ is the conserved charge associated with time and space translation invariance via Noether's theorem:

$$P^\mu = \int d^3x T^{0\mu}(x). \quad (1)$$

Transitioning to quantum mechanics, P^μ becomes an operator. The mass M of a state $|P\rangle$ at rest ($P = (M, \vec{0})$) is given by its rest energy:

$$M = \langle P | \int d^3x \hat{T}^{00}(x) | P \rangle. \quad (2)$$

At this stage, it is crucial to specify the physical nature of M . Derived strictly from spacetime symmetries, this is the *inertial mass*, which quantifies the energy content and the resistance of the system to acceleration. Interestingly, according to Einstein's Equivalence Principle, this identical tensor $\hat{T}^{\mu\nu}$ acts as the source of gravity, seamlessly equating this inertial mass with the active gravitational mass.

To make mathematical sense of Equation (2) yielding a finite, measurable mass M , we must initially rely on macroscopic intuition. We implicitly assume the state $|P\rangle$ represents a localized wave packet $|\psi\rangle = \int d^3p \phi(p) |p\rangle$ normalized to unity ($\langle\psi|\psi\rangle = 1$), or a particle

perfectly confined in a finite physical box. Under this assumption, integrating the local energy density over all space converges cleanly to the total physical mass.

2.2 Plane waves and the local-to-global relation

While a localized wave packet provides an intuitive macroscopic picture, rigorous QFT calculations rely on idealized continuous momentum eigenstates: the plane waves. In position space, a plane wave is described by $\langle x|P\rangle \propto e^{-iP \cdot x}$. Because it extends uniformly over all of infinite space, its spatial integral diverges and it cannot be normalized to 1. Instead, its inner product yields a Dirac distribution:

$$\langle P'|P\rangle = (2\pi)^3 \delta^{(3)}(\vec{P} - \vec{P}'). \quad (3)$$

Using infinite plane waves forces us to rethink our mass definition. We cannot blindly evaluate the spatial integral in Equation (2). Due to translation invariance, the local energy density for a momentum eigenstate is strictly constant across all space: $\langle P|\hat{T}^{00}(x)|P\rangle = \langle P|\hat{T}^{00}(0)|P\rangle$. Consequently, the spatial integral trivially diverges, yielding the infinite macroscopic volume of the universe:

$$\int d^3x \langle P|\hat{T}^{00}(x)|P\rangle = \langle P|\hat{T}^{00}(0)|P\rangle \int d^3x \rightarrow \infty. \quad (4)$$

Instead, a mathematical trick consists in extracting a strictly *local* property—the value of the energy-momentum tensor evaluated exactly at the origin, $\hat{T}^{\mu\nu}(0)$ —from the *global* macroscopic integral $\hat{P}^\mu = \int d^3x \hat{T}^{0\mu}(x)$. To achieve this extraction, we invoke spacetime translation invariance. The local operator at any point x evolves from the origin according to $\hat{T}^{\mu\nu}(x) = e^{i\hat{P} \cdot x} \hat{T}^{\mu\nu}(0) e^{-i\hat{P} \cdot x}$. By evaluating the global momentum operator between two plane waves, the momentum operators act on the external states to produce an exponential phase:

$$\langle P'|\hat{P}^\mu|P\rangle = \int d^3x \langle P'|\hat{T}^{0\mu}(x)|P\rangle = \int d^3x e^{i(P-P') \cdot x} \langle P'|\hat{T}^{0\mu}(0)|P\rangle. \quad (5)$$

Performing the spatial integration over d^3x rigorously yields a Dirac distribution. This factors out the infinite volume of space, isolating the local matrix element evaluated strictly at $x = 0$:

$$\langle P'|\hat{P}^\mu|P\rangle = (2\pi)^3 \delta^{(3)}(\vec{P} - \vec{P}') \langle P|\hat{T}^{0\mu}(0)|P\rangle. \quad (6)$$

On the other hand, the direct action of the momentum operator on its eigenstate ($\hat{P}^\mu|P\rangle = P^\mu|P\rangle$) combined with our plane wave normalization (Equation 3) gives:

$$\langle P'|\hat{P}^\mu|P\rangle = P^\mu \langle P'|P\rangle = P^\mu \left[(2\pi)^3 \delta^{(3)}(\vec{P} - \vec{P}') \right]. \quad (7)$$

By equating Equation (7) with Equation (6) and dividing both sides by the Dirac distribution, we immediately extract the purely local relation: $\langle P|\hat{T}^{0\mu}(0)|P\rangle = P^\mu$. Evaluating the 00-component for a state at rest ($P = (M, \vec{0})$) perfectly recovers the intuitive rest-energy relation $M = \langle P|\hat{T}^{00}(0)|P\rangle$, but it is now completely independent of any spatial integral.

2.3 Relativistic normalization and general tensor structure

However, this non-relativistic normalization is fundamentally inconsistent with the invariant phase space required in QFT. Because of Lorentz contraction, the standard covariant normalization must absorb a kinematic energy factor to remain consistent across all reference frames:

$$\langle P'|P\rangle = (2\pi)^3 2E_P \delta^{(3)}(\vec{P} - \vec{P}'). \quad (8)$$

Inserting this rigorous QFT normalization back into our prior equality yields an extra factor of $2E_P = 2P^0$:

$$P^\mu \left[(2\pi)^3 2E_P \delta^{(3)}(\vec{P} - \vec{P}') \right] = (2\pi)^3 \delta^{(3)}(\vec{P} - \vec{P}') \langle P|\hat{T}^{0\mu}(0)|P\rangle. \quad (9)$$

Dividing by the Dirac distribution, we extract the exact local matrix element for the 0μ components:

$$\langle P|\hat{T}^{0\mu}(0)|P\rangle = 2P^0 P^\mu. \quad (10)$$

To fully capture the internal dynamics of the particle and formulate a covariant definition of mass, we must generalize this result to all components of the symmetric rank-2 tensor $\hat{T}^{\mu\nu}(0)$. The most general form evaluated for a single-particle state must be built from the only available Lorentz-invariant structures: the momenta $P^\mu P^\nu$ and the metric $\eta^{\mu\nu}$. This leads to:

$$\langle P|\hat{T}^{\mu\nu}(0)|P\rangle = 2P^\mu P^\nu A(0) + \eta^{\mu\nu} B(0). \quad (11)$$

Starting from this equation, we can build a coherent definition of the mass of a quantum field as follows.

1. Fixing the kinematic form factor:

For a state at rest ($P^0 = M, P^i = 0$), the pure energy component $\hat{T}^{00}$ must yield $2M^2$ to satisfy our newly derived relation $\langle P|\hat{T}^{00}(0)|P\rangle = 2P^0 P^0$. This natural constraint mathematically fixes the kinematic form factor $A(0) = 1$, leaving us with an unknown pressure term $B(0)$:

$$\langle P|\hat{T}^{\mu\nu}(0)|P\rangle = 2P^\mu P^\nu + \eta^{\mu\nu} B(0). \quad (12)$$

2. Eliminating the pressure term:

Why do we identify $B(0)$ as a pressure term? In classical physics and fluid dynamics, the purely spatial components of the energy-momentum tensor, T^{ij} (with $i, j \in \{1, 2, 3\}$), represent the momentum flux, which corresponds to mechanical stress. For an isotropic system at rest, these diagonal spatial components are exactly equal to the internal pressure p :

$$\hat{T}^{ii} = p. \quad (13)$$

Let us evaluate these spatial components for our quantum state at rest ($P^i = 0$). Using the $(+, -, -, -)$ metric signature, the spatial part of the metric is $\eta^{ii} = -1$. Injecting this into our general tensor structure yields:

$$\langle P|\hat{T}^{ii}(0)|P\rangle = 2(0)(0) + (-1)B(0) = -B(0). \quad (14)$$

Therefore, $B(0)$ is directly proportional to the internal pressure of the particle. According to the von Laue theorem (1911), a stable bound state (like the proton) does not explode nor collapse; its internal forces must perfectly balance. Mathematically, the spatial integral of these internal pressures strictly vanishes:

$$\int d^3x \langle P | \hat{T}^{ii}(x) | P \rangle = 0 \implies B(0) = 0. \quad (15)$$

3. The fundamental trace relation:

With $B(0)$ eliminated, the tensor simplifies exactly to $\langle P | \hat{T}^{\mu\nu}(0) | P \rangle = 2P^\mu P^\nu$. By contracting the indices with $\eta_{\mu\nu}$, we obtain $P^\mu P_\mu = M^2$, revealing a fundamental equivalence between the mass and the Lorentz scalar trace:

$$2M^2 = \langle P | \hat{T}_\mu^\mu(0) | P \rangle. \quad (16)$$

3 The photon case and the trace anomaly

Now that we have established the fundamental kinematic relation $2M^2 = \langle P | \hat{T}_\mu^\mu(0) | P \rangle$ for any quantum state, let us apply it to the simplest gauge field: the photon in QED. Our goal is to understand how quantum fluctuations alter the classical picture, step by step. Note that because we are evaluating local operators between identical momentum eigenstates, exact translation invariance ensures that the spatial dependence completely factors out: $\langle P | \hat{\mathcal{O}}(x) | P \rangle = \langle P | \hat{\mathcal{O}}(0) | P \rangle$. For the remainder of this paper, we will omit the spacetime argument (0) for brevity when computing these matrix elements.

3.1 Scale invariance and the classical trace

Let us temporarily return to the classical theory, where fields are standard functions rather than quantum operators. Classically, the energy-momentum tensor of the free electromagnetic field is:

$$T_{\text{EM}}^{\mu\nu} = F^{\mu\alpha} F_\alpha^\nu - \frac{1}{4} \eta^{\mu\nu} F_{\alpha\beta} F^{\alpha\beta}. \quad (17)$$

In a strictly four-dimensional spacetime, the metric contraction yields $\eta_{\mu\nu} \eta^{\mu\nu} = 4$. Taking the trace of this tensor gives exactly zero:

$$(T_\mu^\mu)_{\text{class}} = F_{\alpha\beta} F^{\alpha\beta} - \frac{1}{4} (4) F_{\alpha\beta} F^{\alpha\beta} = 0. \quad (18)$$

Through our mass definition $2M^2 = \langle \gamma | T_\mu^\mu | \gamma \rangle$, this classical scale invariance dictates that the photon is massless. There is no fundamental length or mass scale in classical electromagnetism.

3.2 Dimensional regularization and the arbitrary scale μ

In QFT, the vacuum is not empty. Quantum fluctuations, such as virtual electron-positron pairs (vacuum polarization), generate mathematical infinities in the loop integrals. To handle these infinities rigorously, physicists use *dimensional regularization* [3]: we analytically continue the spacetime dimensions from 4 to $d = 4 - \epsilon$.

Moving to QFT, the fields become quantum operators (denoted with hats). Moving away from 4 dimensions has a profound consequence on the dimensions of the fields and couplings. In natural units ($\hbar = c = 1$), the action $S = \int d^d x \hat{\mathcal{L}}$ must be strictly dimensionless. Because the volume element $d^d x$ has a mass dimension of $-d$, the Lagrangian density $\hat{\mathcal{L}}$ must have a mass dimension of $+d$. Consequently, to keep the interaction term between electrons and photons dimensionless in d dimensions, the bare coupling constant of the underlying theory, denoted e_0 , can no longer be dimensionless. It must acquire a mass dimension. We extract this dimension by introducing an *arbitrary energy scale* μ :

$$[e_0] = \mu^{\epsilon/2}. \quad (19)$$

Furthermore, in d dimensions, the metric contraction becomes $\eta_{\mu\nu}\eta^{\mu\nu} = d = 4 - \epsilon$. The trace of the energy-momentum operator is no longer zero, but acquires a purely dimensional term involving the bare field strength operator $\hat{F}_0$:

$$\hat{T}_\mu^\mu = -\hat{F}_{0\alpha\beta}\hat{F}_0^{\alpha\beta} + \frac{d}{4}\hat{F}_{0\alpha\beta}\hat{F}_0^{\alpha\beta} = -\frac{\epsilon}{4}\hat{F}_{0\alpha\beta}\hat{F}_0^{\alpha\beta}. \quad (20)$$

Naively, as we remove the regulator ($\epsilon \rightarrow 0$), one might expect this trace to vanish. However, the quantum vacuum has a hidden trick.

3.3 The renormalization constant Z_e

Writing e_0 solely as a dimensional shift of a physical coupling $e(\mu)$ is an incomplete picture. The bare coupling e_0 must also absorb all the mathematical infinities (the $1/\epsilon$ poles) generated by the quantum loops (vacuum polarization and vertex corrections). To achieve this, we introduce a dimensionless renormalization constant Z_e . The complete, true relation linking the bare, infinite coupling of the unobservable Lagrangian to the physical, renormalized coupling $e(\mu)$ measured in experiments is:

$$e_0 = \mu^{\epsilon/2} Z_e e(\mu). \quad (21)$$

Here, $e(\mu)$ is finite, while Z_e contains all the ultraviolet divergences.

3.4 Independence of the Bare Coupling and the β Function

Equation (21) encodes a geometric constraint. Indeed, the bare coupling e_0 is the fundamental parameter of the universe at an infinite energy cutoff; it is completely oblivious to the arbitrary scale μ that human physicists chose for dimensional regularization. Therefore, the derivative of e_0 with respect to μ must be strictly zero. This physical necessity forces a non-trivial evolution on the renormalized coupling $e(\mu)$, known as the running of the coupling:

$$\mu \frac{de_0}{d\mu} = 0 \implies \mu \frac{d}{d\mu} \left(\mu^{\epsilon/2} Z_e e(\mu) \right) = 0. \quad (22)$$

Applying the chain rule, we expand this exact differential:

$$\frac{\epsilon}{2} Z_e e(\mu) + \mu \frac{dZ_e}{d\mu} e(\mu) + Z_e \mu \frac{de(\mu)}{d\mu} = 0. \quad (23)$$

The term $\mu \frac{de}{d\mu}$ is precisely the definition of the Callan-Symanzik β function, $\beta(e)$ [4]. By isolating it, we explicitly extract the β function entirely in terms of the renormalization constant Z_e :

$$\beta(e) = -e \left(\frac{\epsilon}{2} + \frac{\mu}{Z_e} \frac{dZ_e}{d\mu} \right). \quad (24)$$

In the physical limit where $\epsilon \rightarrow 0$, we find that $\beta(e) = -e \frac{d \ln Z_e}{d \ln \mu}$. The running of the coupling is the direct, unavoidable consequence of the scale-independence of the bare theory.

3.5 Renormalization and the emergence of the anomaly

We must now relate the bare mathematical field operator $\hat{F}_0$ in Equation (20) to the physical, renormalized operator $\hat{F}$ that we observe. They are related by the field renormalization constant Z_3 (where $\hat{F}_0 = \sqrt{Z_3} \hat{F}$). In the minimal subtraction scheme, Z_3 absorbs the infinities generated by the vacuum polarization loops. It contains a divergent pole in $1/\epsilon$ that is precisely proportional to the β function:

$$Z_3 = 1 - \frac{2\beta(e)}{e\epsilon} + \mathcal{O}(\beta^2). \quad (25)$$

Let us substitute $(\hat{F}_0)^2 = Z_3 \hat{F}^2$ into our d -dimensional trace:

$$\hat{T}_\mu^\mu = -\frac{\epsilon}{4} \left(1 - \frac{2\beta(e)}{e\epsilon} \right) \hat{F}^2 = -\frac{\epsilon}{4} \hat{F}^2 + \frac{\beta(e)}{2e} \hat{F}^2. \quad (26)$$

This equation is the core of the phenomenon. As we return to the physical four-dimensional world by taking the limit $\epsilon \rightarrow 0$, the first term smoothly vanishes. However, the second term contains the product of a vanishing kinematic dimension (ϵ) and a quantum divergence ($1/\epsilon$). They cancel each other perfectly, leaving a finite, non-zero remainder. This explicitly breaks scale invariance and produces the so-called *trace anomaly* [5]:

$$\hat{T}_\mu^\mu = \frac{\beta(e)}{2e} \hat{F}_{\alpha\beta} \hat{F}^{\alpha\beta}. \quad (27)$$

3.6 Evaluating the anomaly: normal ordering and the field operator

To evaluate the trace anomaly on a single free photon state $|\gamma(P)\rangle$, we must be rigorous about how we define the quantum operators. While the Time-Ordered product ($T\{\dots\}$) is used to compute Feynman propagators and scattering amplitudes, defining physical observables like the energy-momentum tensor requires *normal ordering* (denoted $:\hat{\mathcal{O}}:$). Let us recall the origin of this requirement: the quantum vacuum is teeming with zero-point fluctuations. If we do not normal-order the operators, the expectation value of the vacuum energy ($\langle 0|\hat{T}^{00}|0\rangle$) would diverge to infinity. Normal ordering systematically forces all creation operators ($\hat{a}^\dagger$) to the left of annihilation operators ($\hat{a}$), ensuring that the pure vacuum expectation value is strictly zero by definition.

To see this in action, we explicitly write the quantized electromagnetic field operator in its plane-wave expansion:

$$\hat{A}_\mu(x) = \int \frac{d^3k}{(2\pi)^3 2E_k} \sum_\lambda \left[\varepsilon_\mu(k, \lambda) \hat{a}_{\vec{k}, \lambda} e^{-ik \cdot x} + \varepsilon_\mu^*(k, \lambda) \hat{a}_{\vec{k}, \lambda}^\dagger e^{ik \cdot x} \right]. \quad (28)$$

The field is naturally split into an annihilation part $\hat{A}_\mu^{(+)} \propto \hat{a}$ (which destroys a photon) and a creation part $\hat{A}_\mu^{(-)} \propto \hat{a}^\dagger$ (which creates a photon). Consequently, the field strength tensor $\hat{F}_{\alpha\beta} = \partial_\alpha \hat{A}_\beta - \partial_\beta \hat{A}_\alpha$ splits analogously into $\hat{F}_{\alpha\beta}^{(+)}$ and $\hat{F}_{\alpha\beta}^{(-)}$.

3.7 Matrix element computation

We must evaluate the normal-ordered square of the field tensor: $:\hat{F}^2: \equiv :\hat{F}_{\alpha\beta}\hat{F}^{\alpha\beta}:$. Expanding the square into its creation and annihilation components, and applying normal ordering (moving all $\hat{F}^{(-)}$ to the left), yields:

$$:\hat{F}^2: = \hat{F}_{\alpha\beta}^{(-)}\hat{F}^{(-)\alpha\beta} + 2\hat{F}_{\alpha\beta}^{(-)}\hat{F}^{(+)\alpha\beta} + \hat{F}_{\alpha\beta}^{(+)}\hat{F}^{(+)\alpha\beta}. \quad (29)$$

Now, we sandwich this operator between our initial and final single-photon states, rigorously keeping the quantum expectation brackets: $\langle\gamma(P)|:\hat{F}^2:|\gamma(P)\rangle$.

- The term $\hat{F}_{\alpha\beta}^{(+)}\hat{F}^{(+)\alpha\beta}$ attempts to annihilate two photons from a 1-photon state. It yields exactly 0.
- The term $\hat{F}_{\alpha\beta}^{(-)}\hat{F}^{(-)\alpha\beta}$ creates two photons, turning the ket into an orthogonal 3-photon state. Its inner product with the 1-photon bra $\langle\gamma(P)|$ is exactly 0.

Only the cross-term survives. The annihilation operator $\hat{F}^{(+)}$ acts to the right to destroy the initial photon, extracting its momentum P and polarization ε . Symmetrically, the creation operator $\hat{F}^{(-)}$ acts to the left on the bra. Thus, the exact quantum expectation value reduces to:

$$\langle\gamma(P)|:\hat{F}^2:|\gamma(P)\rangle = 2\langle\gamma(P)|\hat{F}_{\alpha\beta}^{(-)}(x)\hat{F}^{(+)\alpha\beta}(x)|\gamma(P)\rangle. \quad (30)$$

Taking the derivatives $\partial_\alpha(e^{\mp i P \cdot x}) = \mp i P_\alpha e^{\mp i P \cdot x}$, the action of the operators on the states pulls down the momenta:

$$\hat{F}^{(+)\alpha\beta}(x)|\gamma(P)\rangle = -i(P^\alpha \varepsilon^\beta - P^\beta \varepsilon^\alpha)e^{-i P \cdot x}|0\rangle, \quad (31)$$

$$\langle\gamma(P)|\hat{F}_{\alpha\beta}^{(-)}(x) = \langle 0|i(P_\alpha \varepsilon_\beta^* - P_\beta \varepsilon_\alpha^*)e^{i P \cdot x}. \quad (32)$$

Substituting these into our expectation value, the plane-wave exponentials ($e^{i P x}e^{-i P x}$) perfectly cancel out. The vacuum inner product yields $\langle 0|0\rangle = 1$. Expanding the product of the terms in parentheses gives:

$$\begin{aligned} \langle\gamma|:\hat{F}^2:|\gamma\rangle &= 2(-i)(i) \left[P_\alpha \varepsilon_\beta^* P^\alpha \varepsilon^\beta - P_\alpha \varepsilon_\beta^* P^\beta \varepsilon^\alpha - P_\beta \varepsilon_\alpha^* P^\alpha \varepsilon^\beta + P_\beta \varepsilon_\alpha^* P^\beta \varepsilon^\alpha \right] \\ &= 2 \left[P^2(\varepsilon^* \cdot \varepsilon) - (P \cdot \varepsilon^*)(P \cdot \varepsilon) - (P \cdot \varepsilon^*)(P \cdot \varepsilon) + P^2(\varepsilon^* \cdot \varepsilon) \right]. \end{aligned} \quad (33)$$

Grouping the terms, we arrive at the exact analytical result for the matrix element:

$$\langle\gamma|:\hat{F}^2:|\gamma\rangle = 4 \left[P^2(\varepsilon^* \cdot \varepsilon) - |P \cdot \varepsilon|^2 \right]. \quad (34)$$

3.8 Transversality and the on-shell condition

To finalize this computation, we must impose the physical constraints of the theory. The equation above still contains the unphysical longitudinal and time-like degrees of freedom of the polarization vector ε . The fundamental principle of QED is $U(1)$ gauge invariance, meaning the physics must remain invariant under the transformation $A_\mu \rightarrow A_\mu + \partial_\mu \alpha$. In momentum space, this implies that any physical observable must remain strictly invariant if we shift the polarization vector by the momentum: $\varepsilon_\mu \rightarrow \varepsilon_\mu + cP_\mu$. To ensure this invariance (and to satisfy the Lorenz gauge condition $\partial_\mu A^\mu = 0$), the photon must couple exclusively to a conserved current. This directly imposes the *Ward Identity* on physical states, which acts as a transversality condition: the momentum of the photon must be strictly orthogonal to its polarization vector:

$$P \cdot \varepsilon = 0. \quad (35)$$

Furthermore, we apply the general on-shell condition for a free particle, $P^2 = M^2$. Let us note that at this level, we obviously do not assume that the photon mass is zero. Substituting these two constraints ($P \cdot \varepsilon = 0$ and $P^2 = M^2$) back into our expectation value, the second term vanishes completely, leaving:

$$\langle \gamma | : \hat{F}^2 : | \gamma \rangle = 4M^2(\varepsilon^* \cdot \varepsilon). \quad (36)$$

Since the physical spacelike polarization vector is normalized such that $(\varepsilon^* \cdot \varepsilon) = -1$, the expectation value of the squared field tensor is strictly proportional to the squared mass of the state:

$$\langle \gamma | : \hat{F}^2 : | \gamma \rangle \propto M^2. \quad (37)$$

3.9 The mass gap equation for the photon

We now have two independent ways to evaluate the expectation value of the trace $\langle \hat{T}_\mu^\mu \rangle$. Equating our pure kinematic definition of mass ($2M^2$) with the dynamical result of the trace anomaly evaluated on the photon state ($\propto \beta(e)M^2$), we obtain the *mass gap equation*:

$$2M^2 = c \cdot \beta(e)M^2 \implies M^2 \left(1 - \frac{c}{2} \cdot \beta(e) \right) = 0. \quad (38)$$

This is the decisive conclusion for QED. Since the loop correction governed by the $\beta(e)$ function is a relatively small quantum effect, it cannot exactly cancel the pure kinematic constant 1. The only mathematically rigorous solution to this equation is strictly $M = 0$. Finally, we find that the photon is massless even in the presence of the trace anomaly in QFT.

3.10 The deep structure of QED: the Ward identity

Before turning to QCD, it is crucial to understand why the photon trace anomaly behaves so beautifully and avoids generating mass. The fundamental renormalization of QED involves three constants: Z_1 for the electron-photon vertex, Z_2 for the electron wavefunction,

and Z_3 for the photon field. The charge renormalization constant Z_e we introduced earlier is a combination of these:

$$Z_e = Z_1 Z_2^{-1} Z_3^{-1/2}. \quad (39)$$

However, the $U(1)$ gauge symmetry of QED rigorously enforces the Ward-Takahashi identity, which mandates that the vertex and fermion corrections exactly cancel each other: $Z_1 = Z_2$ [6]. This simplification leads to:

$$Z_e \sqrt{Z_3} = 1. \quad (40)$$

Physically, this relation means that because the photon itself carries no electric charge, it cannot directly alter the fundamental charge of the electron. The mathematical probability of an electron emitting a virtual photon (the vertex correction) is perfectly balanced by the correction to its own free propagation (wavefunction renormalization). Therefore, the physical vacuum can only screen the electric charge like a macroscopic dielectric medium—a phenomenon entirely and exclusively captured by the vacuum polarization Z_3 . There is a perfect mapping between the trace anomaly and the photon propagator. As we will see in the next sections, this elegant simplicity breaks down in QCD due to gluon self-interactions, setting the stage for the generation of mass.

3.11 Non-linear QED: the Light-by-Light contribution

To further test the robustness of our mass derivation, we must consider extreme physical conditions where Maxwell's equations become non-linear. In effective field theories, such as the Euler-Heisenberg Lagrangian describing Light-by-Light (LbL) scattering [7], the electromagnetic action acquires non-linear quantum corrections due to the integration of heavy fermion loops. The complete leading-order effective Lagrangian takes the form:

$$\hat{\mathcal{L}}_{\text{eff}} = -\frac{1}{4} \hat{F}_{\alpha\beta} \hat{F}^{\alpha\beta} + c_1 (\hat{F}_{\alpha\beta} \hat{F}^{\alpha\beta})^2 + c_2 (\hat{F}_{\alpha\beta} \tilde{\hat{F}}^{\alpha\beta})^2. \quad (41)$$

Here, $\tilde{\hat{F}}^{\alpha\beta} = \frac{1}{2} \epsilon^{\alpha\beta\rho\sigma} \hat{F}_{\rho\sigma}$ is the dual field strength tensor operator, and the coupling constants c_1 and c_2 are strictly dimensionful ($c_1, c_2 \propto \alpha^2/m_e^4$).

Because these constants carry a mass dimension of -4 (due to the electron mass m_e being integrated out), this non-linear Lagrangian explicitly breaks scale invariance at the classical level. It is crucial to distinguish this from the trace anomaly: here, the scale symmetry is explicitly broken by the massive parameters of the effective theory, rather than by quantum fluctuations of a scale-invariant bare theory. Let us compute the exact macroscopic energy-momentum tensor operator for this full non-linear theory. Using the standard metric variation $\hat{T}_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta(\sqrt{-g} \hat{\mathcal{L}}_{\text{eff}})}{\delta g^{\mu\nu}}$, the trace of the energy-momentum tensor in four dimensions ($\eta_{\mu\nu} \eta^{\mu\nu} = 4$) yields a striking result:

$$(\hat{T}_\mu^\mu)_{\text{LbL}} = -4 \left[c_1 (\hat{F}_{\alpha\beta} \hat{F}^{\alpha\beta})^2 + c_2 (\hat{F}_{\alpha\beta} \tilde{\hat{F}}^{\alpha\beta})^2 \right]. \quad (42)$$

Unlike the classical free photon, the trace of the macroscopic non-linear field is *not zero*. Does this mean that an intense electromagnetic field generates a rest mass for the photon?

To answer this, we must evaluate the normal-ordered operator $: \hat{T}_\mu^\mu :$ on our single free photon state $|\gamma(P)\rangle$:

$$2M_{\text{LbL}}^2 = -4\langle\gamma(P)| : [c_1(\hat{F}^2)^2 + c_2(\hat{F}\tilde{\hat{F}})^2] : |\gamma(P)\rangle. \quad (43)$$

This is where the algebraic structure of the quantum Fock space is essential. Both operators $(\hat{F}^2)^2$ and $(\hat{F}\tilde{\hat{F}})^2$ are products of four local field tensors. When decomposed into creation $(\hat{F}^{(-)} \propto \hat{a}^\dagger)$ and annihilation $(\hat{F}^{(+)} \propto \hat{a})$ operators and strictly normal-ordered, they contain terms combining four operators (e.g., $a^\dagger a^\dagger a^\dagger a^\dagger$, $a^\dagger a^\dagger a a$, etc.). For the expectation value $\langle\gamma|\dots|\gamma\rangle$ to be non-zero, the operator must destroy exactly the same number of photons it creates. The only term in a normal-ordered 4-field operator capable of conserving the photon number is the crossed term proportional to two creations and two annihilations:

$$: \hat{F}^4 : \text{ or } : (\hat{F}\tilde{\hat{F}})^2 : \supset \hat{F}^{(-)}\hat{F}^{(-)}\hat{F}^{(+)}\hat{F}^{(+)} \propto \hat{a}^\dagger\hat{a}^\dagger\hat{a}\hat{a}. \quad (44)$$

Let us carefully apply this operator to our single-photon state $|\gamma(P)\rangle$:

$$(\hat{a}^\dagger\hat{a}^\dagger\hat{a}\hat{a})|1 \text{ photon}\rangle = (\hat{a}^\dagger\hat{a}^\dagger\hat{a})|0\rangle = 0. \quad (45)$$

Because the state contains only one photon, the second annihilation operator strikes the empty vacuum, yielding exactly zero. All other terms in the normal-ordered expansion (which change the photon number) are trivially orthogonal to the bra $\langle\gamma|$. We obtain:

$$\langle\gamma| : \hat{T}_\mu^\mu : |\gamma\rangle \equiv 0 \implies M = 0. \quad (46)$$

This result is much more profound than a simple kinematic impossibility (such as Furry's theorem forbidding a photon to split into two, $\gamma \not\rightarrow \gamma\gamma$). It reveals the fundamental difference between mass and interactions. Mass is an intrinsic property of a single quantum state. However, the breaking of scale invariance in non-linear QED is an *emergent, collective* property. The non-linear operators inherently require at least two initial photons to activate the Light-by-Light scattering, $\gamma\gamma \rightarrow \gamma\gamma$. A single photon traveling through the vacuum is the minimal quantum of energy; it cannot polarize the vacuum all by itself. Thus, the single-particle mass gap equation remains exactly zero, and the free photon strictly retains its masslessness despite the non-linear dynamics of the vacuum.

4 QCD and the origin of the proton mass

4.1 The QCD Lagrangian and the chiral limit

To understand the origin of the proton mass, we must begin with the full Quantum Chromodynamics (QCD) Lagrangian, which describes both the gluon fields and the quarks as quantum operators:

$$\hat{\mathcal{L}}_{\text{QCD}} = -\frac{1}{4}\hat{G}_{\mu\nu}^a\hat{G}^{a\mu\nu} + \sum_q \bar{\psi}_q(i\not{D} - m_q)\hat{\psi}_q. \quad (47)$$

Here, D_μ is the gauge-covariant derivative, m_q represents the bare quark masses, and $\hat{G}_{\mu\nu}^a$ is the non-Abelian field strength tensor operator. Unlike photons, gluons carry color charge and strongly self-interact. This non-linear behavior is encoded in the tensor:

$$\hat{G}_{\mu\nu}^a = \partial_\mu \hat{A}_\nu^a - \partial_\nu \hat{A}_\mu^a + gf^{abc} \hat{A}_\mu^b \hat{A}_\nu^c. \quad (48)$$

When we evaluate the trace of the energy-momentum operator for this complete theory, it contains two distinct contributions: the explicit symmetry breaking from the quark masses, and the quantum trace anomaly from the gluons [5, 8]:

$$\hat{T}_\mu^\mu = \frac{\beta(g)}{2g} \hat{G}_{\alpha\beta}^a \hat{G}_a^{\alpha\beta} + \sum_q m_q (1 + \gamma_m) \bar{\psi}_q \hat{\psi}_q. \quad (49)$$

where γ_m is the anomalous dimension of the quark mass operator.

Why do we often neglect the second term? In the real physical world, the masses of the up (u) and down (d) quarks that constitute the proton are merely a few MeV, which is overwhelmingly negligible compared to the total proton mass of ≈ 938 MeV. The heavier quarks are systematically integrated out. This allows us to work in the chiral limit ($m_q \rightarrow 0$). In this limit, the classical theory is perfectly scale-invariant, and the entire mass of the proton is dynamically generated by the purely gluonic trace anomaly:

$$\hat{T}_\mu^\mu \approx \frac{\beta(g)}{2g} \hat{G}_{\alpha\beta}^a \hat{G}_a^{\alpha\beta}. \quad (50)$$

Applying our fundamental kinematic relation ($2M^2 = \langle P | \hat{T}_\mu^\mu | P \rangle$) to the proton state $|P\rangle$, we find that the proton mass is essentially the expectation value of this anomaly:

$$2M_p^2 \approx \langle P | \frac{\beta(g)}{2g} \hat{G}^2 | P \rangle. \quad (51)$$

4.2 The failure of the plane wave approximation

We established that for a free photon, the mathematical evaluation of $\langle \gamma | \hat{F}^2 | \gamma \rangle$ vanishes perfectly. Why doesn't the exact same mathematical cancellation apply to gluons? The answer lies in the fundamental difference between QED and QCD: the sign of the β function. In QED, $\beta(e) > 0$, meaning the theory is weakly coupled at large macroscopic distances (infrared). This allows photons to exist as free asymptotic states. We can rigorously describe them with simple, transverse plane waves ($\varepsilon_\mu e^{-ik \cdot x}$), which guarantees the exact factorization of the momentum tensor and forces the mass to vanish $\langle \hat{F}^2 \rangle \propto P^2 \varepsilon^2 - (P \cdot \varepsilon)^2 = 0$.

In stark contrast, QCD features asymptotic freedom: $\beta(g) < 0$ [9]. The coupling diverges at large distances, leading to absolute color confinement [10]. Gluons cannot exist as free macroscopic plane waves. Instead, we evaluate the anomaly on the proton $|P\rangle$, which is a composite bound state. Inside the proton, gluons are continuously exchanging momentum. They are strongly interacting, *virtual particles*. Because they are off-shell, their internal momenta k do not satisfy the free-space condition $k^2 = 0$. The exact algebraic cancellation $k^2 \varepsilon^2 - (k \cdot \varepsilon)^2 = 0$ is therefore mathematically impossible to achieve for a confined gluon. Consequently, the internal dynamics never cancel out, and the expectation value of the gluon field squared remains strictly non-zero: $\langle \hat{G}^2 \rangle \neq 0$.

4.3 The QED Landau pole

To fully understand why QED leaves the photon massless while QCD generates the proton mass, we must mathematically solve the Callan-Symanzik equation. For QED, the β function is positive, $\mu \frac{\partial e}{\partial \mu} = \beta_0 e^3$ with $\beta_0 > 0$. Separating the variables to integrate from a low-energy reference scale μ_0 (where the coupling is e_0) to an arbitrary scale μ :

$$\int_{e_0}^{e(\mu)} \frac{de}{e^3} = \int_{\mu_0}^{\mu} \beta_0 \frac{d\mu'}{\mu'}. \quad (52)$$

Performing the integration yields:

$$-\frac{1}{2e^2(\mu)} + \frac{1}{2e_0^2} = \beta_0 \ln \left(\frac{\mu}{\mu_0} \right) \implies e^2(\mu) = \frac{e_0^2}{1 - 2\beta_0 e_0^2 \ln \left(\frac{\mu}{\mu_0} \right)}. \quad (53)$$

This equation reveals a profound singularity. As the energy scale μ increases, the denominator shrinks. At a specific, extremely high energy scale called the *Landau Pole* (Λ_{LP}), the denominator vanishes and the coupling diverges to infinity:

$$1 - 2\beta_0 e_0^2 \ln \left(\frac{\Lambda_{\text{LP}}}{\mu_0} \right) = 0 \implies \Lambda_{\text{LP}} = \mu_0 \exp \left(\frac{1}{2\beta_0 e_0^2} \right). \quad (54)$$

Because $\beta_0 e_0^2$ is very small in QED, this pole is pushed to exponentially unreachable energies. At any accessible macroscopic scale, the coupling remains extremely weak, photons do not self-interact, and the plane-wave approximation holds perfectly. The generated mass scale is totally decoupled from our reality, leaving $M_\gamma = 0$.

4.4 Dimensional transmutation and the 1 GeV scale

In QCD, the β function is strictly negative: $\mu \frac{\partial g}{\partial \mu} \approx -\beta_0 g^3$. Solving this exact same differential equation with a minus sign reverses the physical picture. The strong coupling g inevitably grows as we probe larger distances (lower energies μ). Eventually, the strong interaction becomes so overwhelmingly powerful that perturbation theory breaks down completely, permanently trapping color charges inside hadrons. Because of this absolute confinement, gluons cannot radiate away into the vacuum; their energy remains strictly localized. Solving this differential equation translates this geometric running of the coupling into a fundamental, invariant physical mass scale Λ_{QCD} [11]:

$$\Lambda_{\text{QCD}} = \mu \exp \left(-\frac{1}{2\beta_0 g^2(\mu)} \right) \approx 250 \text{ MeV}. \quad (55)$$

Because of confinement, the internal virtual gluon dynamics must scale with the only available dimensionful parameter in the chiral limit. The mass gap equation for QCD equates our kinematic definition with this dynamical scale:

$$2M_p^2 = C \cdot \Lambda_{\text{QCD}}^2 \implies M_p \propto \Lambda_{\text{QCD}} \approx 1 \text{ GeV}. \quad (56)$$

The exact proportionality constant C arises from the complex, non-perturbative multi-gluon dynamics. This confirms that the macroscopic mass of hadrons is dynamically generated by the trace anomaly.

4.5 The three ways of a Gauge Theory

This analysis reveals that the macroscopic physical spectrum of any gauge theory is entirely dictated by the sign of its β function:

- **The QED case** ($\beta > 0$): The coupling diverges in the deep ultraviolet (Landau Pole). The generated mass scale is pushed to unreachable energies. At low energies, particles remain strictly free, the plane-wave approximation holds, and the anomaly evaluates to zero ($M_\gamma = 0$).
- **The conformal case** ($\beta = 0$): The coupling is strictly constant. No dimensional transmutation occurs, exact scale invariance is maintained ($\hat{T}_\mu^\mu = 0$), and all particles remain strictly massless.
- **The QCD case** ($\beta < 0$): The coupling diverges in the infrared, defining the confinement scale Λ_{QCD} . Free plane waves are forbidden. The trace anomaly condenses into bound states, breaking scale invariance macroscopically and generating mass ($M_p \sim 1 \text{ GeV}$).

4.6 The rigidity of gauge symmetry and proton structure

It is worth noting that expanding the gluon kinetic term $\hat{\mathcal{L}}_{\text{gluon}} = -\frac{1}{4}\hat{G}^2$ explicitly reveals the origin of this complexity. It yields a free kinetic term alongside cubic (3-gluon) and quartic (4-gluon) interaction vertices. The rigidity of gauge invariance imposes a strict mathematical constraint: the exact same coupling g completely determines both the 3-point and 4-point vertices. When experiments at the LHC or HERA perform QCD fits on splitting functions at Leading Order and Next-to-Leading Order, they are directly validating this exact, geometric non-Abelian correlation that ultimately traps the gluons and generates the proton mass.

4.7 Non-linear QCD: gluon recombination and saturation

To complete our parallel with the non-linear Light-by-Light scattering in QED, we must address the extreme non-linear regime of QCD. How exactly do non-linearities prevent the trace anomaly $\langle P|\hat{G}^2|P\rangle$ from diverging to infinity? To answer this, we must recognize that the expectation value $\langle P|\hat{G}^2|P\rangle$ is deeply related to the integral of the gluon number density inside the proton. At high energies (or small momentum fractions x), gluons copiously radiate lower-energy gluons via bremsstrahlung ($g \rightarrow gg$). If we rely solely on the linear evolution equations of QCD (such as BFKL or DGLAP), this splitting cascade causes the gluon density $\rho(x)$ to grow indefinitely as a power law: $\rho(x) \propto x^{-\lambda}$. If we were to integrate this unchecked exponential growth to calculate the total energy-momentum trace, the proton mass would trivially diverge to infinity.

However, the non-Abelian nature of QCD provides a natural, geometric regulatory mechanism: *gluon recombination* ($gg \rightarrow g$). The proton has a finite transverse size (roughly 1 fm). As the number of gluons explodes, their wave functions eventually

overlap. At this critical density, the non-linear interaction vertices we identified earlier ($g^2 f^{abc} f^{ade} \hat{A}^4$) become overwhelmingly dominant. The evolution of the gluon density transitions from a linear growth to a non-linear dynamic equation. In the Color Glass Condensate (CGC) effective theory, this is governed by the Balitsky-Kovchegov (BK) equation [12], whose structural form is essentially a quantum logistic equation:

$$\frac{\partial \rho}{\partial Y} \approx \alpha_s \rho - \alpha_s^2 \rho^2. \quad (57)$$

Here, $Y \sim \ln(1/x)$ is the rapidity. The first term ($+\alpha_s \rho$) represents the linear gluon splitting that drives the density toward infinity. The crucial second term ($-\alpha_s^2 \rho^2$) represents the non-linear recombination.

Mathematically, this non-linear negative feedback forces the growth to stop. The gluon density completely plateaus (saturates) at a dynamically generated emergent scale, the saturation momentum Q_s [13]. Does this extreme non-linear saturation alter our mass gap equation? No, it secures it. In Section 3.11, we saw that non-linear operators like $:\hat{F}^4:$ evaluated to zero on a single-photon state because a single photon traversing the vacuum cannot self-interact, preserving $M_\gamma = 0$. In QCD, however, the proton $|P\rangle$ is intrinsically a macroscopic, collective bound state containing a saturated bath of interacting virtual gluons. The non-linear recombination does not create a new rest mass out of nowhere; rather, it tames the infrared divergences of the theory. Because the gluon density plateaus rather than diverges, the integral over this distribution remains strictly finite. Thus, non-linear recombination ensures that the expectation value of the trace anomaly $\langle P|\hat{G}^2|P\rangle$ does not blow up. It forces the anomaly to remain a finite, well-behaved macroscopic observable. We reach a beautiful physical symmetry: while QED non-linearities are merely a high-energy perturbation transparent to single free particles, QCD non-linearities (saturation) act as the fundamental mathematical stabilizer of the proton mass.

5 Discussion: Cosmology, curved spacetime and energy conservation

A profound physical question arises when situating this microscopic framework within our macroscopic expanding universe. In Section 2, our entire mathematical derivation of the mass gap relied heavily on global translation invariance ($\partial_\mu \hat{T}^{\mu\nu} = 0$), which allowed us to define the four-momentum as a conserved charge: $\hat{P}^\mu = \int d^3x \hat{T}^{0\mu}$. However, global time-translation invariance is explicitly broken in cosmology. Does this cosmological reality invalidate our quantum derivation of the proton mass?

5.1 The cosmological breakdown of global energy

To state the paradox clearly, let us place our quantum fields in the macroscopic background of the Friedmann-Lemaître-Robertson-Walker (FLRW) metric, which describes a

homogeneous, isotropic, expanding universe [14]:

$$ds^2 = dt^2 - a^2(t)d\vec{x}^2, \quad (58)$$

where $a(t)$ is the cosmological scale factor. In General Relativity, the standard partial derivative of Noether's theorem is replaced by the covariant derivative: $\nabla_\mu \hat{T}^{\mu\nu} = 0$. Expanding this covariant derivative using the Christoffel symbols (Γ) yields:

$$\partial_\mu \hat{T}^{\mu\nu} + \Gamma_{\mu\alpha}^\mu \hat{T}^{\alpha\nu} + \Gamma_{\mu\alpha}^\nu \hat{T}^{\mu\alpha} = 0. \quad (59)$$

Let us evaluate the temporal component ($\nu = 0$) for a macroscopic perfect fluid at rest. The energy density is $\rho = \langle \hat{T}^{00} \rangle$ and the isotropic pressure is $p = \langle \hat{T}^{ii} \rangle$. The non-zero Christoffel symbols for the FLRW metric yield $\Gamma_{0i}^i = \dot{a}/a$, which is precisely the Hubble parameter $H(t)$. The temporal equation simplifies to the fundamental continuity equation:

$$\dot{\rho} + 3\frac{\dot{a}}{a}(\rho + p) = 0. \quad (60)$$

In Section 2, we defined the mass (rest energy) via a spatial volume integral. In the FLRW metric, the proper physical volume element is not simply d^3x , but $dV = \sqrt{-g} d^3x = a^3(t)d^3x$. Multiplying the continuity equation by this volume a^3 , we can rewrite it as:

$$\frac{d}{dt}(\rho a^3) = -p \frac{d}{dt}(a^3) \implies dE = -p dV. \quad (61)$$

This confirms the macroscopic reality: the total energy $E = \rho V$ contained in a co-moving cosmological volume is *not conserved*. As the universe expands ($dV > 0$), a fluid with positive pressure performs mechanical work against the expanding spacetime fabric, losing energy. Because $\hat{P}^0$ is no longer a constant of motion, the global plane waves defined in Equation (7) are technically not exact solutions of the universe.

5.2 Local Poincaré symmetry and Riemann normal coordinates

Does the failure of global energy conservation ruin the trace anomaly derivation of the proton mass? No, because of the strict locality of the quantum operators and the Equivalence Principle. While spacetime is macroscopically curved, General Relativity guarantees that at any arbitrary spacetime point x_0 (such as the center of a proton), we can always construct a local inertial frame known as Riemann normal coordinates. In this specific coordinate system, the metric behaves locally as flat Minkowski spacetime, with corrections appearing only at the second order in distance [15]:

$$g_{\mu\nu}(x) = \eta_{\mu\nu} - \frac{1}{3}R_{\mu\alpha\nu\beta}(x_0)x^\alpha x^\beta + \mathcal{O}(x^3), \quad (62)$$

where $R_{\mu\alpha\nu\beta}$ is the Riemann curvature tensor evaluated at the origin.

This mathematical expansion is crucial. If we evaluate the physics strictly at the origin and neglect the $\mathcal{O}(x^2)$ curvature terms, the local connection coefficients (Christoffel symbols) strictly vanish. The covariant derivative reduces exactly to the flat-space partial derivative ($\nabla_\mu \rightarrow \partial_\mu$), and local energy conservation ($\partial_\mu \hat{T}^{\mu\nu} = 0$) is perfectly restored.

This seamlessly validates the exact mathematical setup we relied upon in Sections 2 to 4. Because the proton is a highly localized bound state (with a physical radius $r_p \approx 0.84$ fm), the quantum field operators $\hat{G}^2$ only interact within this microscopic region where spacetime is effectively flat.

5.3 Separation of scales and the modified trace anomaly

To rigorously quantify this flat-space approximation, we must examine how the trace anomaly itself is modified by curved spacetime. When computing QFT loop integrals in a curved background, dimensional regularization generates additional divergent terms proportional to the geometric invariants of the metric. The complete trace anomaly for a gauge theory in curved spacetime takes the form [15]:

$$\hat{T}_\mu^\mu = \frac{\beta(g)}{2g} \hat{G}^2 + c_1 R + c_2 \left(R_{\mu\nu} R^{\mu\nu} - \frac{1}{3} R^2 \right) + \dots \quad (63)$$

Notice a critical notational subtlety in this equation: while the energy-momentum tensor and the gluon field strength retain their operator hats ($\hat{T}$, $\hat{G}$) as quantized fields, the curvature scalars (R , $R_{\mu\nu}$) do not. This reflects the framework of semiclassical gravity (or QFT in curved spacetime). Because we lack a full theory of quantum gravity, the macroscopic spacetime metric $g_{\mu\nu}$ is treated as a classical background field, while the Standard Model gauge fields living upon it are fully quantized.

We can now mathematically evaluate the separation of scales. The macroscopic curvature of the universe is determined by the Hubble constant $H_0 \approx 1.5 \times 10^{-42}$ GeV. The Ricci scalar is of the order of $R \sim H_0^2 \sim 10^{-84}$ GeV². On the other hand, the purely internal QCD dynamics of the proton are governed by the confinement scale $\Lambda_{\text{QCD}} \approx 0.25$ GeV, leading to a trace anomaly density of order $\Lambda_{\text{QCD}}^4 \sim 10^{-2}$ GeV⁴.

When evaluating the matrix element $2M_p^2 = \langle P | \hat{T}_\mu^\mu | P \rangle$, the relative contribution of the cosmological expansion to the proton mass is governed by the dimensionless ratio:

$$\frac{\text{Cosmological correction}}{\text{QCD dynamics}} \propto \left(\frac{H_0}{\Lambda_{\text{QCD}}} \right)^2 \sim 10^{-82}. \quad (64)$$

This ratio is unimaginably small. The timescale of internal gluon dynamics is around 10^{-24} seconds, completely decoupled from the cosmological Hubble time of 10^{17} seconds.

Therefore, while global translation invariance is broken, the local Poincaré symmetry protecting the matrix elements in Section 2 is restored to an exquisite precision ($1 - 10^{-82}$). The proton mass is an intrinsic, purely local quantum property. The expanding universe stretches the macroscopic space between galaxy clusters, diluting their global energy density, but space itself does not expand inside the proton.

6 Conclusion

The origin of mass is a dynamic, collective phenomenon. The rigorous evaluation of the trace anomaly demonstrates that a gauge theory's physical spectrum is entirely dic-

tated by the sign of its β function and its underlying non-linear dynamics. Through step-by-step matrix element evaluations, we have shown that while the photon remains a massless mirage—immune to its own vacuum non-linearities because it is an asymptotic free state—the proton mass is a direct macroscopic manifestation of color confinement and gluon saturation. The non-linear self-interactions of QCD are not merely corrections; they are the fundamental mechanism that generates the Λ_{QCD} scale and prevents the trace anomaly from diverging. Finally, this intricate quantum framework is securely grounded in local spacetime symmetries that survive even in a dynamic, expanding cosmological background.

References

- [1] P. W. Higgs, Phys. Rev. Lett. **13**, 508 (1964); F. Englert and R. Brout, Phys. Rev. Lett. **13**, 321 (1964).
- [2] C. G. Callan, S. Coleman, and R. Jackiw, Annals Phys. **59**, 42-73 (1970).
- [3] G. 't Hooft and M. Veltman, Nucl. Phys. B **44**, 189-213 (1972).
- [4] C. G. Callan, Phys. Rev. D **2**, 1541 (1970); K. Symanzik, Commun. Math. Phys. **18**, 227 (1970).
- [5] R. J. Crewther, Phys. Rev. Lett. **28**, 1421 (1972).
- [6] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory*, Addison-Wesley (1995).
- [7] W. Heisenberg and H. Euler, Z. Phys. **98**, 714 (1936).
- [8] X. Ji, Phys. Rev. Lett. **74**, 1071 (1995).
- [9] D. J. Gross and F. Wilczek, Phys. Rev. Lett. **30**, 1343 (1973); H. D. Politzer, Phys. Rev. Lett. **30**, 1346 (1973).
- [10] G. 't Hooft, Nucl. Phys. B **72**, 461 (1974).
- [11] C. D. Roberts, Symmetry **12**, 1468 (2020); Craig D. Roberts and Sebastian M. Schmidt, Eur. Phys. J. ST **229**, 3319 (2020).
- [12] I. Balitsky, Nucl. Phys. B **463**, 99-160 (1996); Y. V. Kovchegov, Phys. Rev. D **60**, 034008 (1999).
- [13] L. V. Gribov, E. M. Levin, and M. G. Ryskin, Phys. Rept. **100**, 1-150 (1983); L. McLerran and R. Venugopalan, Phys. Rev. D **49**, 2233 (1994).
- [14] S. M. Carroll, *Spacetime and Geometry: An Introduction to General Relativity*, Addison-Wesley (2004).

- [15] N. D. Birrell and P. C. W. Davies, *Quantum Fields in Curved Space*, Cambridge University Press (1982).